Post Assignment IV

4. Function in C

Q.1. What is a function? Why we use function in C language? Give an example

→ A function is block of code that performs a specific task and can be execute by calling it from other part of the program.

function helps in orgainizing code, making it modular are reuseable. Thus function is use in C language.

ex.

```
# include <stdio.h>
int add(int a, int b);
int main()
{
  int result = add(3,5);      // function
  printF("Sum: %d\n", result);
  return 0;
}
int add (a, b)
{
  return a+b;
}
```

Q.2 Distinguish between Library function and User defined function in C and explain with example.

→

User defined function	Library function
1. A programmer create a function according to the requirement of a program which is called User-defined function.	1. A function whose prototype are already defined in the C library is called the library function
2. A Userdefined function does not require writing any code inside the header file	2. All the library function are predefined inside the header file.
3. A Userdefined function is required to write the complete code before using the function in a program	3. We don't require writing a complete code to use the library function in a program

4. {int n1, n2, sum;
n1 = 10; n2 = 20;
sum = addnum(n1, n2);
Printf("sum=%d", sum);
} int addnum(int a, int b)
{sum = a+b;
return sum;
}
	# include <stdio.h>
	# include <math.h>
	int main()
	{ int a=125, squareroot;
	squareroot = sqrt(a);
	printf("%d", squareroot);
	return 0;
	}

Q.3 Write a program to generate Fibonacci Serise using recursive function

→
```
# include<stdio.h>
int Fibonacci (int F)
{
 int a=0,b=1,c;
 iF(F>0)
 {
  c = a +b;
  a = b;
  b = c;
  printF("%d", c);
  fibonacci(F-1);
  return 0;
 }}
int main()
{
 int F;
 printF("Enter number");
 ScanF("%d", &F);
 printF("The fibonacci Serise is:");
 printF("%d %d" 0, 1);
 Fibonacci(F -2);
 return 0;
 }
```